

Touchless Doorbell For The Post-Covid World

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This low-cost, simple, reliable, and effective circuit for a doorbell works without touching. It can be quite useful in our post-Covid-19 world as it minimises the spread of virus through doorbells.

Circuit and working

The author's prototype is shown in Fig. 1. Fig. 2 shows the block diagram of the touchless doorbell circuit while Fig. 3 shows the complete circuit. The circuit is built around dual timer NE556 (IC1), transistor BC547 (T1), IR receiver TSOP1738 (IRRX1), IR transmitter LED (IRTX1), 5.1V Zener diode (ZD1), rectifier diode (D1), and 12V single-changeover relay (RL1).

NE556 is a dual NE555 timer IC capable of producing precise time delays or oscillation. It has two independent timers. One timer is configured as an astable multivibrator (built around pins 8 through 13), which is used to drive the IR sensor circuit, and the other as a monostable multivibrator (built around pins 1 through 6) to drive the relay circuit.

The output pin 5 of IC1 is used to drive the relay, which extends AC power to the doorbell. The whole circuit is powered by a 12V DC power supply, which can be an SMPS based power supply with current rating of 500mA, or any other regulated power supply.

The circuit operates as a proximity sensor and detects nearby objects without any physical contact. When you place your hand or finger near infrared (IR) LED IRTX1, its IR light is reflected back and received by the nearby placed TSOP1738 sensor (IRRX1). As shown in block diagram of the sensor in Fig. 4, it has a PIN diode, pre-amplifier, band-pass filter, and demodulator. The epoxy package is designed as an IR filter.

The IR transmitting LED (IRTX1) is connected to output pin 9 of the



Fig. 1: Author's prototype

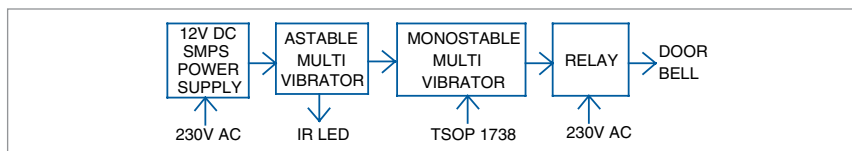


Fig. 2: Block diagram

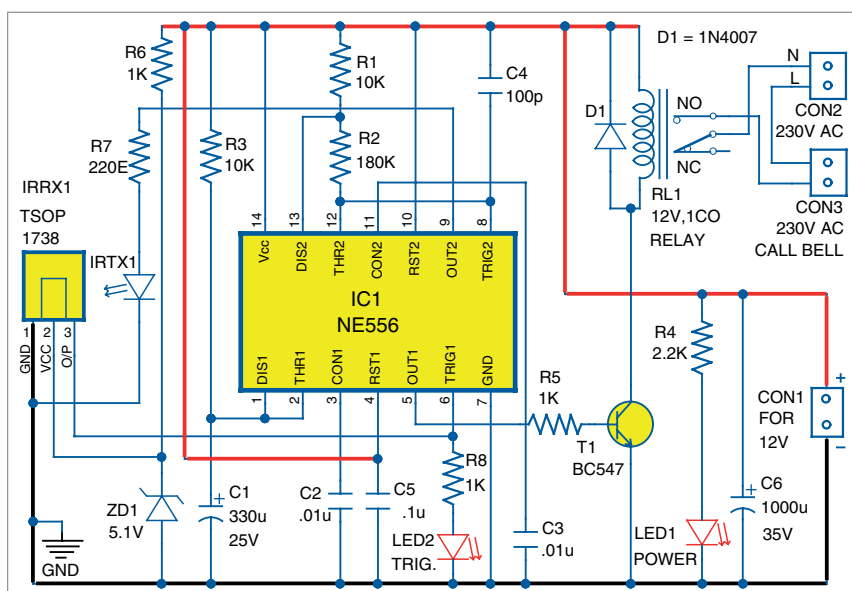


Fig. 3: Circuit diagram of touchless doorbell

first timer section of NE556 (IC1). This multivibrator generates the frequency in the range of 32kHz to 38kHz, which is the operating frequency of the TSOP1738

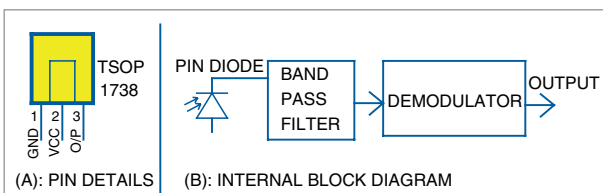


Fig. 4: Block diagram of TSOP1738